

Azure CDN

Azure **Content Delivery Network (CDN)** is a globally distributed service designed to accelerate content delivery, reduce latency, and improve performance for users accessing web applications, APIs, media, and other online assets. It caches content at **edge servers** closest to end users, reducing the load on the original source.

How Azure CDN Works

1. A user requests content (e.g., an image or webpage).
2. If cached, the nearest **CDN edge server** delivers the content instantly.
3. If not cached, the CDN fetches it from the origin (Azure Storage, Web Apps, or an external server), caches it, and then delivers it to the user.
4. Future requests are served directly from the cache, improving speed and reducing bandwidth costs.

Key Features of Azure CDN

1. Global Content Distribution

- Azure CDN has **multiple Points of Presence (PoPs)** worldwide, ensuring content is closer to users, improving **performance and reliability**.

2. Performance Optimization

- **Caching static & dynamic content:** Speeds up image, video, CSS, and JavaScript delivery.
- **Compression:** Uses **Gzip & Brotli** to reduce file sizes and enhance page load speed.
- **Adaptive Streaming:** Improves media streaming by adjusting quality based on network conditions.

3. Security Enhancements

- **DDoS Protection:** Defends against Distributed Denial of Service attacks.
- **Custom HTTPS & TLS Encryption:** Ensures secure content delivery.
- **Token Authentication:** Restricts content access to authorized users.

4. Scalability & High Availability

- **Load Balancing:** Distributes traffic across multiple CDN nodes to prevent overload.
- **Failover Support:** If an edge server is down, requests are rerouted to the next best server.

5. Deep Integration with Azure Services

- Works seamlessly with **Azure Blob Storage, Web Apps, API Management, and Media Services**.

- Supports integration with third-party services and on-premises environments.

Use Cases of Azure CDN

1. Website & Web Application Acceleration

A global e-commerce website can use Azure CDN to **load product images, stylesheets, and JavaScript faster**, ensuring a smooth shopping experience.

2. Media Streaming & Video Delivery

A video streaming platform can use Azure CDN for **adaptive bitrate streaming**, reducing buffering and improving video playback quality.

3. Software Downloads & Updates

Companies providing software updates can distribute large files using Azure CDN, ensuring faster, reliable downloads worldwide.

4. API & Web Services Optimization

A financial service with a **REST API** can cache responses using Azure CDN, reducing server load and improving response times for global users.

5. Online Gaming & Mobile Apps

Online multiplayer games use Azure CDN for **quick delivery of assets, skins, maps, and patches**, reducing lag and enhancing player experience.

Benefits of Azure CDN

- **Faster Content Delivery** – Reduces latency by caching content closer to users.
- **Lower Bandwidth Costs** – Reduces load on origin servers, cutting data transfer costs.
- **Better User Experience** – Faster websites, smoother streaming, and instant downloads.
- **Scalability** – Handles traffic spikes and millions of simultaneous users.
- **Enhanced Security** – Protects against cyber threats with DDoS mitigation and secure authentication.
- **Global Reach** – Ensures high availability across continents with Azure's global network.

Azure CDN Pricing

Azure CDN pricing depends on **data transfer and requests**. It offers **three CDN options**:

1. **Azure CDN from Microsoft** – Integrated with Azure services.
2. **Azure CDN from Verizon** – Enhanced security and acceleration.
3. **Azure CDN from Akamai** – Best for dynamic site acceleration.

Pricing varies based on **region, traffic, and CDN provider**. You can estimate costs using the **Azure Pricing Calculator**.

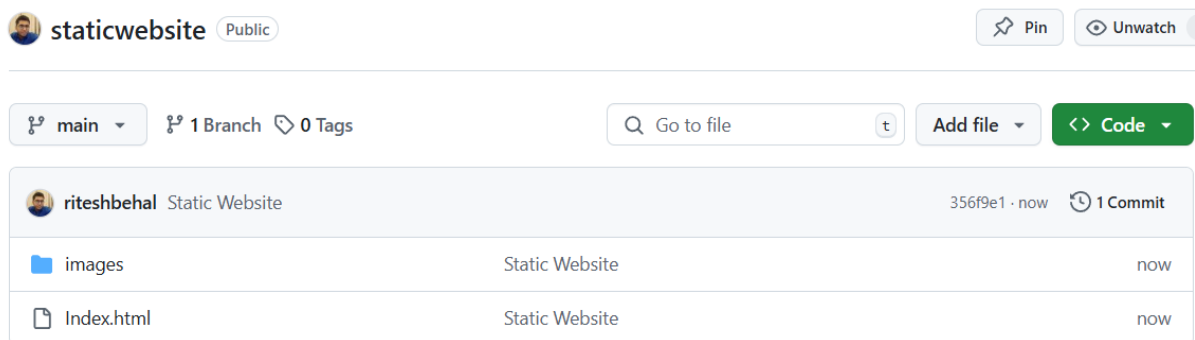
Conclusion

Azure CDN is an essential service for businesses requiring **fast, secure, and scalable content delivery**. Whether for websites, streaming platforms, gaming, or APIs, Azure CDN enhances user experience while **reducing latency and infrastructure costs**.

The process involves hosting a static website on GitHub and deploying it using Azure Static Web Apps. First, a GitHub repository is created, and a static website is uploaded. In Azure, a static web app is created using the free tier, linked to the GitHub repository, and deployed with West Europe as the staging environment. Next, an Azure CDN profile is set up via Front Door, with the static web app's DNS as the origin. The CDN endpoint reduces load time by caching content closer to users. The end goal is to improve website performance and global accessibility.

😄 To begin with the Lab

1. In this lab first we will create a new GitHub Repository and upload a static website onto it. You can download the Static website attached to the lab and then upload it.



2. Now in the Azure Portal search for static website from the marketplace. Then choose your Resource Group and for the Hosting plan choose free tier.

Create Static Web App ...

Basics Deployment configuration Advanced Tags Review + create

App Service Static Web Apps is a streamlined, highly efficient solution to take your static app from source code to global high availability. Pre-rendered content is distributed globally with no web servers required. [Learn more](#)

Project Details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ MSDN Platforms Subscription ✓

Resource Group * ⓘ NewRG ✓

[Create new](#)

Static Web App details

Name * staticwebapp121 ✓

Hosting plan

The hosting plan dictates your bandwidth, custom domain, storage, and other available features. [Compare plans](#)

Plan type

☒ Free: For hobby or personal projects

☐ Standard: For general purpose production apps

☐ Dedicated (preview): For general purpose production apps

3. After that choose GitHub for your source and connect your GitHub account with Azure then choose your organization and repository. In the end choose HTML for the build preset. Click on next.

Deployment details

Source

☒ GitHub ☐ Azure DevOps ☐ Other

GitHub account

Pulkit-Kumar-0 [Change account](#) ⓘ

i If you can't find an organization or repository, you might need to enable additional permissions on GitHub. You must have write access to your chosen repository to deploy with GitHub Actions. ✕

Organization *

Pulkit-Kumar-0

Repository *

CoffeeJunkie

Branch *

main

Build Details

Enter values to create a GitHub Actions workflow file for build and release. You can modify the workflow file later in your GitHub repository.

Build Presets

 HTML

i These fields will reflect the app type's default project structure. Change the values to suit your app. [Learn more](#) ↗

4. In the advanced tab you have to choose West Europe for your staging environment.

Basics Deployment configuration **Advanced** Tags Review + create

Azure Static Web Apps provide global high availability for your applications' production environment. Static Web Apps also comes with the option to have a managed Azure Functions API. Managed functions and non-production environments for your Static Web Apps resource are regional. [Learn more](#) ↗

Azure Functions and staging details

Region for Azure Functions API and staging environments *

West Europe

Enterprise-grade edge

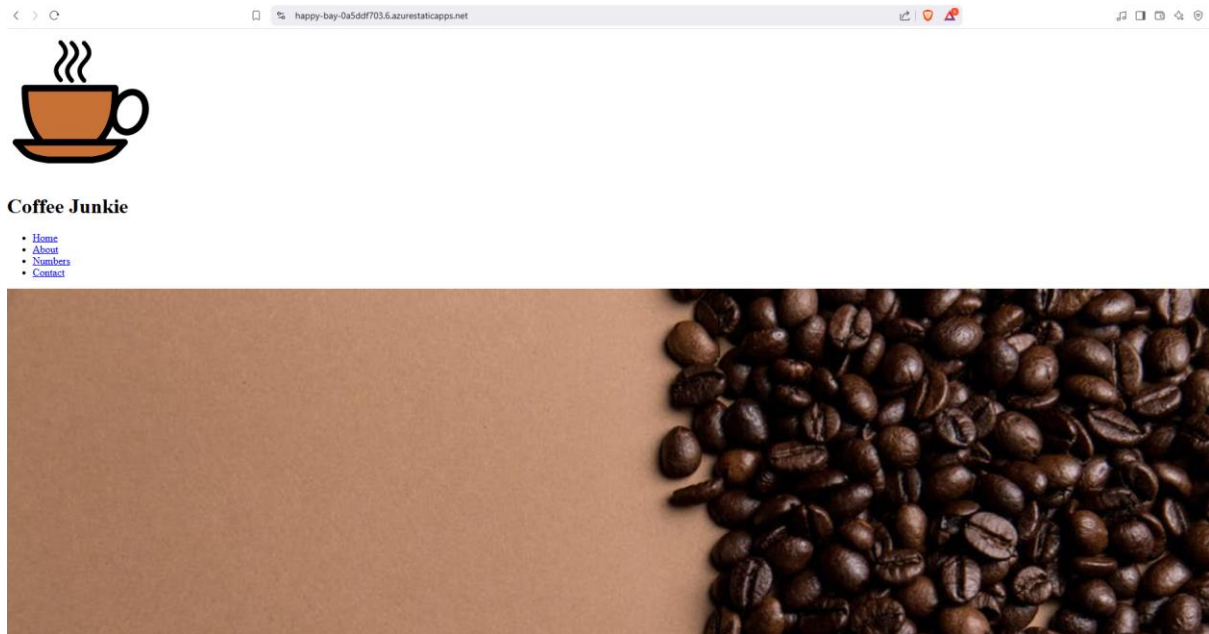
Extend this Static Web App with enterprise-grade edge powered by Azure Front Door. This seamless integration will keep your traffic on the best path to your app, improve website speed, and increase throughput for your global users with edge load balancing across 118+ points of presence, SSL offload, and application acceleration. [Learn more](#) ↗

Enterprise-grade edge

☐

i Enterprise-grade edge requires a standard hosting plan. [Change hosting plan](#) ↗

5. Move to the review page and create your static web app. Once your static web app is created copy the DNS name and paste it into a new tab you will see the index page.



6. Now we need to create the CDN service. In the marketplace search for front door and CDN profiles. Click on create.

Front Door and CDN profiles ...

Microsoft



Front Door and CDN profiles [Add to Favorites](#)

Microsoft | Azure Service

★ 2.9 (10 ratings)

Plan

Front Door and CDN profiles 

Create

7. In the offerings choose explore other offerings and then select Azure CDN Standard. Click on Continue.

Compare offerings ...

Microsoft Azure

Choose between Azure Front Door and other offerings.

Azure Front Door ☐

Azure Front Door is a secure cloud CDN which provides static and dynamic content acceleration, global load balancing and protection of your apps, APIs and websites with intelligent threat protection.

Explore other offerings ☒

See offerings for our Azure Front Door (classic) and Azure CDN Standard from Microsoft (classic), along with our partner offerings.

Choose other offerings

Azure Front Door (classic) ☐

A global and scalable entry point that uses Microsoft global network to provide dynamic application acceleration, load balancing and security.

i Azure Front Door (classic) will be retired on March 31, 2027. It is highly recommended to start new creations on Azure Front Door Standard and Premium to avoid future migrations and benefit from all the new and enhanced capabilities.

Azure CDN Standard from Microsoft (classic) ☒

A global content delivery network that uses Microsoft global network for content caching and acceleration.

i Azure CDN Standard from Microsoft (classic) will be retired on September 31, 2027. It is highly recommended to start new creations on Azure Front Door Standard and Premium to avoid future migrations and benefit from all the new and enhanced capabilities.

Continue

8. Now you need to choose your resource group then give a unique name.

Subscription * MSDN Platforms Subscription ▼

Resource group * NewRG ▼
[Create new](#)

Resource group region ⓘ North Europe ▼

Profile details

Name * cdnprofile121 ✓

Region Global
 ⓘ CDN profiles are global resources that work across Azure regions

Pricing tier * Microsoft CDN (classic) ▼
[View full pricing details](#) ↗

9. For the endpoint settings you need to provide an endpoint name then for the origin type choose custom origin and in the hostname put the DNS name of your Static webapp.

Endpoint settings

Create a new CDN endpoint ☒

CDN endpoint name * staticwebsite1 ✓
 .azureedge.net

Origin type * Custom origin ▼

Origin hostname * ⓘ happy-bay-0a5ddf703.6.azurestaticapps.net ✓

Query string caching behavior * ⓘ Ignore query strings ▼

10. Here you can see that the CDN profile has been created now you can inside your endpoint and take the endpoint hostname.

cdnprofile121 Front Door and CDN profile

Search Endpoint Purge Move Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Resource group (move) : [NewRG](#)

Status : Active

Subscription (move) : [MSDN Platforms Subscription](#)

Subscription ID : d6549a66-c45c-4979-840c-3b356da446b0

Pricing Tier : Microsoft CDN (classic)

Endpoints

Hostname	Status	Protocol	Origin type	Custom domains
staticwebsite1.azureedge.net	Running	HTTP, HTTPS	Custom origin	

[JSON View](#)

Azure CDN Standard from Microsoft (classic) will be retired on September 30, 2027. Migrate to [AFD Standard and Premium](#) to avoid any service disruption and benefit from all new and enhanced capabilities. [Learn more](#)

staticwebsite1 (cdnprofile121/staticwebsite1) ✖ ☆ ...

Endpoint

Search

+ Custom domain Purge Stop Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

Azure CDN Standard from Microsoft (classic) will be retired on September 30, 2027. Migrate to [AFD Standard and Premium](#) to avoid any service disruption and benefit from all new and enhanced capabilities. [Learn more](#)

Essentials [JSON View](#)

Resource group (move) : [NewRG](#)

Status : Running

Location : Global

Subscription (move) : [MSDN Platforms Subscription](#)

Subscription ID : d6549a66-c45c-4979-840c-3b356da446b0

Endpoint hostname : <https://staticwebsite1.azureedge.net>

Origin hostname : <https://happy-bay-0a5ddf703.6.azurestaticapps.net>

Protocols : HTTP, HTTPS

Optimization type : General web delivery

Custom domains

Hostname ↑↓ Custom HTTPS ↑↓ Details ↑↓

There are no custom domains to display

11. Here you can see that our using the endpoint from the CDN profile is taking less time to load our website.

staticwebsite1.azureedge.net



Coffee Junkie

- Home
- About
- Numbers
- Contact



Network

Filter

Preserve log Disable cache No throttling

Y Filter

Fetch/XHR Doc CSS JS Font Img Media Manifest WS Wasm Other

Name	Status	Type	Initiator	Size	Time
staticwebsite1.azureedge.net	200	document	Other	2.9 kB	51 ms
logo.png	200	png	index01	3.8 kB	44 ms
background.jpeg	200	jpeg	index01	80.4 kB	128 ms
house-blend.jpeg	200	jpeg	index04	13.1 kB	83 ms
french-roast.jpeg	200	jpeg	index02	17.5 kB	89 ms
columbian-roast.jpeg	200	jpeg	index02	13.3 kB	93 ms

6 requests 131 kB transferred 132 kB resources Finish: 203 ms DOMContentLoaded: 69 ms